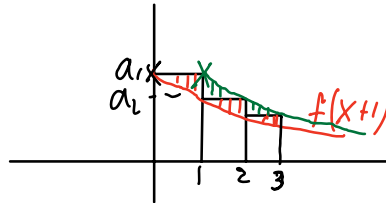


שעת קבלה

$$\int_1^\infty f(x) dx \leq \sum_{n=1}^\infty a_n \leq a_1 + \int_1^\infty f(x) dx$$

$f(x) = \frac{1}{x}$
 $\forall n \in \mathbb{N}$

$a_1 + a_2 + a_3 + \dots + \sum_{n=4}^\infty a_n$



$$\int_0^\infty f(x+1) dx = \int_1^\infty f(x) dx$$

$$a_n = a_1 \cdot q^n$$

הקשר $|q| < 1$

הגבול בקצרה $a_n \rightarrow 0$

$$\sum_{n=1}^\infty \frac{1}{n^p}$$

$p > 1$
הקשר

$$a_n = b_n - b_{n+1}$$

$$\sum_{n=1}^\infty a_n = b_1 - \lim_{n \rightarrow \infty} b_n$$

$$b_n$$

$$b_n < a_n$$

$$\sum a_n \leq \sum b_n$$

הקשר $L = \lim_{n \rightarrow \infty} \frac{b_n}{a_n}$

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$$a_n < b_n$$

$$\sum b_n$$

$$\sum a_n$$

$$s_n = b_1 - b_{n+1}$$

$$L = \lim_{n \rightarrow \infty} \frac{b_n}{a_n}$$

$\sum b_n, \sum a_n, 0 < L < \infty$
הקשר

$$\frac{b_n}{L-\epsilon} < a_n < b_n \left(\frac{1}{L+\epsilon} \right)$$

$$(L-\epsilon)a_n < b_n < a_n(L+\epsilon)$$

$$\exists N: \forall n > N$$

$$\underbrace{\frac{\cos x - \cos(2x)}{1}} + \frac{\cos(2x) - \cos(3x)}{2} + \dots$$

$$\forall \varepsilon > 0 \quad \exists N: \forall n, m > N \quad |a_n - a_m| < \varepsilon$$

$$|S_n - S_m| = \left| \sum_{k=m+1}^n a_k \right| < \varepsilon$$

$n > m$ 故

$$a_k = \frac{\cos(kx) - \cos((k+1)x)}{k}$$

$$\left| \frac{\cos((n+1)x) - \cos((n+2)x)}{n+1} + \frac{\cos((n+2)x) - \cos((n+3)x)}{n+2} + \dots \right|$$

$$\left| \frac{\cos((n+1)x)}{n+1} - \frac{\cos((n+1)x)}{n} + \cos((n+2)x) \left(\frac{1}{n+2} - \frac{1}{n+1} \right) + \dots \right| \leq$$

$$\leq \left| \frac{\cos((n+1)x)}{n+1} \right| + \left| \frac{\cos((n+1)x)}{n} \right| + \underbrace{\left| \frac{1}{n+2} - \frac{1}{n+1} \right| + \left| \frac{1}{n+3} - \frac{1}{n+2} \right| + \dots}_{\left(\frac{1}{n+1} - \frac{1}{n+2} \right) + \left(\frac{1}{n+2} - \frac{1}{n+3} \right) + \dots}$$

$$\leq \frac{1}{n+1} + \frac{1}{n} + \frac{1}{n+1} - \frac{1}{n+1} \leq \frac{3}{N} < \varepsilon$$

$$N = \left\lceil \frac{3}{\varepsilon} \right\rceil$$

$$\frac{1}{\sqrt{1 \cdot 2}} + \dots + \frac{1}{\sqrt{n(n+1)}}$$

$$S_n - S_n > \varepsilon > 0$$

$$\frac{n(N)}{m(N)}$$

$$N < n$$

$$> \frac{n}{2n+1} > \frac{1}{3}$$

$$S_{2n} - S_n = a_{n+1} + \dots + a_{2n} = \frac{1}{\sqrt{(n+1)(n+2)}} + \frac{1}{\sqrt{(n+2)(n+3)}} + \dots + \frac{1}{\sqrt{(2n)(2n+1)}} > \frac{1}{2n+1}$$

$$\frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n}} = \frac{\sqrt{n} - 1}{\sqrt{n(n+1)}}$$

$$\frac{\sqrt{n}}{\sqrt{n+1}} - \frac{\sqrt{n-1}}{\sqrt{n}} = \frac{n - \sqrt{(n+1)(n-1)}}{\sqrt{n(n+1)}}$$

$$\frac{\sqrt{n+1}}{\sqrt{n}} - \frac{\sqrt{n}}{\sqrt{n+1}} = \frac{1}{\sqrt{n(n+1)}}$$